

Çağatay Gültekin

Software Engineer · Backend (Python/Java) · Machine Learning

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About Me

Software Engineer focused on scalable backend systems, APIs, and data-intensive applications. Designs and implements Python and Java/Spring services and React/TypeScript web applications, including microservices, database architectures, automation workflows, and secure system-to-system integrations. Applies machine learning and data processing experience to build systems integrating data pipelines and ML/LLM components.

Experience

Machine Learning Research Engineer Mar 2025 – Sep 2025

accu:rate GmbH · Crowd Simulation Software & AI · Munich, Germany

• Designed and implemented a modular end-to-end experimentation pipeline in Python, integrating crowd:it simulations with SWIM-based neural surrogate models.

• Built reproducible, configuration-driven experimentation workflows to automate data preprocessing, model training, train/validation/test evaluation, and systematic benchmarking across configurable model architectures.

• Achieved up to 99% prediction accuracy using 17,410 simulation runs, and reached 97% accuracy with only 9 carefully sampled runs using randomized quasi-Monte Carlo sampling.

• Completed the project as an industry-based master's thesis, collaborating with academic and company stakeholders to align the ML approach with production-oriented crowd simulation workflows.

Software Engineer Sep 2024 – Feb 2025

Legalian GmbH · LegalTech & RegTech (KYC/AML) · Munich, Germany

• Designed and implemented a FastAPI microservice in Python, integrating with the Danish Business Register via secure system-to-system API access, including authentication, authorization, and structured error handling.

• Designed and implemented a normalized MySQL database for structured company registry data, including current and historical ownership records, and integrated Elasticsearch for advanced search, querying, and filtering.

• Built data ingestion and retrieval workflows using Pandas to normalize complex CVR API responses and Selenium to retrieve PDFs for records unavailable through the API while handling anti-automation constraints.

• Delivered the CVR microservice as part of an interdisciplinary project, owning its design, implementation, testing, documentation, and Docker-based deployment while ensuring reproducible deployment environments.

Software Engineer Apr 2024 – Sep 2024

itestra GmbH · IT Consulting & Enterprise Software Modernization · Munich, Germany

• Implemented a Java-based AST static analysis engine using Eclipse JDT to detect performance anti-patterns in Java enterprise systems, particularly excessive repository/database calls inside loops.

• Designed call graph analysis and multi-level call-chain tracing to reduce false negatives across indirect call paths and produce structured analysis results with severity levels.

• Extended the analyzer with snapshot-based runtime validation and VisualVM profiling, and integrated it into IntelliJ via a Kotlin plugin for in-IDE execution.

• Delivered the solution as part of a technology consulting project using sprint-based development, client checkpoints, PR-based workflows, CI-driven regression testing, and structured issue tracking.

Graduate Student Research Assistant Feb 2023 – Aug 2023

Helmholtz Munich / TUM · Biomedical AI, Graph ML & Computational Biology · Munich, Germany

• Supported chemCPA's Python/PyTorch codebase by reproducing user-reported issues across model and data workflows and translating findings into technical solutions for the development team.

• Contributed to a joint Helmholtz-TUM research project using chemCPA, a deep learning framework for predicting drug-induced cellular responses, and graph neural network-based molecular representations.

Software Engineer Intern Jul 2021 – Sep 2021

Yapı Kredi Teknoloji · Banking & FinTech · Kocaeli, Türkiye

Built a full-stack web app using React and Angular with a Java/Spring Boot backend, integrating the Atlassian Jira API to auto-generate sprint newsletters for agile teams. Also contributed to PL/SQL automation tools.

IT Assistant Intern Aug 2019 – Sep 2019
Zorluteks Tekstil · Textile Manufacturing & Enterprise IT · Kırklareli, Türkiye
Developed a C#/.NET and SQL-based desktop application in Visual Studio to streamline material updates and database operations. Also migrated spreadsheet data into company databases.

Selected Projects

Legal AI Data Labelling Platform Mar 2024 – Jun 2024
Worked with Legalian GmbH on an internal legal document processing and annotation platform with a React/TypeScript frontend and Flask backend, supporting OCR-based text extraction and ML-assisted classification. Developed reusable UI components following Figma-based workflows for document/ZIP upload, OCR-result review, and label validation. Implemented frontend-backend API contracts, JWT-secured gateway integration, unit and integration testing, and a reproducible Docker setup.

C++/MPI High-Performance Simulation Project Oct 2025 – Mar 2026
Extended a distributed-memory C++ Navier–Stokes solver (NS-EOF) with MPI-based domain decomposition and PETSc integration. Integrated parallel HDF5 output via the HDF5 C API with XDMF metadata for ParaView, replacing fragmented VTK/ASCII output and achieving ∼50% faster I/O and ∼50% smaller file sizes. The HDF5 pipeline is now actively used by a TUM mechanical engineering chair.

Understanding Linear Regions of ReLU Networks
Authored and presented a seminar paper on ReLU linear regions, robustness, expressivity, and over-parameterization under Prof. Dr. Stephan Günnemann at TUM.

Education

Technical University of Munich (TUM) Oct 2022 – Mar 2026
M.Sc. Informatics (Computer Science), degree requirements completed in Oct 2025

Middle East Technical University (METU/ODTÜ) Aug 2016 – Feb 2022
B.Sc. Computer Engineering

Language Skills

Turkish (Native), English (C1/C2 Proficient), German (B2 Upper-Intermediate), French (A2 Basic), Spanish (A2 Basic)

Digital Skills

Core Languages: Python, Java, SQL, C/C++, JavaScript/TypeScript, Kotlin, C#, R, MATLAB

Backend/APIs: FastAPI, Spring Boot, Flask, Django, REST APIs, AuthN/AuthZ, JWT, Hibernate/ORM

Data & Databases: MySQL, PostgreSQL, Oracle, Microsoft SQL Server, MongoDB, SQLite, PL/SQL, Elasticsearch

Machine Learning: Scikit-learn, PyTorch, TensorFlow, NumPy, SciPy, Pandas, Matplotlib, Seaborn

Software Engineering/Tools: React, Angular, Microservices, LLM APIs, RAG, Git, GitHub Actions, CI/CD, Docker, Kubernetes, Maven, Gradle, npm, JUnit, Postman, Selenium, OCR, YAML, Figma

Quality/Performance: Software testing, unit/integration/regression testing, static & dynamic code analysis, call graph construction, performance optimization, profiling, VisualVM, abstract syntax trees

Workflows: GitHub, GitLab, Jira, Bitbucket, Confluence, Trello, Agile workflow, UML, EER modeling, code review

Hobbies & Interests

Vocals and electric guitar · performed with bands in Türkiye, Cyprus, and Munich, including at Gasteig HP8
Dance & Sports · intermediate in salsa and bachata; football, basketball, and volleyball; former licensed table tennis player

References

Prof. Dr. Felix Dietrich · Technical University of Munich · felix.dietrich@tum.de

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